

Total No. of Questions : 8]

SEAT No. :

PC1963

[6353]-297

[Total No. of Pages : 2

T.E. (Mechanical/Automobile)
HONORS IN ELECTRIC VEHICLES
e-Vehicle Technology
(2019 Pattern) (Semester-I) (302031 MJ)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 Q.7, or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw the neat sketches wherever necessary.

- Q1)** a) Write a short note on li-ion battery with its working principle? [4]
b) What are the different chemistries of li-ion batteries? Make its comparative analysis on basis of power, energy and lifespan? [8]
c) Explain the advantages and disadvantages of li-ion batteries? [8]

OR

- Q2)** a) Explain the following parameters of li-ion batteries in short (any two). [4]
i) C-rate
ii) E-rate
iii) Storage capacity
b) What is battery cell balancing? Explain the Issues and remedies for battery balancing. [8]
c) What are the effects of overcharging and termination voltage accuracy on pack capacity of li-ion battery? Explain with suitable graph. [8]

- Q3)** a) Explain nickel-bromide batteries with advantages, disadvantages and applications. [8]
b) Explain lead acid batteries with advantages, disadvantages and applications [8]

OR

- Q4)** a) Explain Nickel-Metal Hydride Batteries with advantages, disadvantages and applications. [8]
b) Explain Li-ion supercapaciter with advantages, disadvantages and applications. [8]

P.T.O.

- Q5) a)** What do you mean by drive system in electric vehicles? Explain with its significance for manufacturers and drivers. [8]
- b)** Write in short different types of motors used in electric vehicle? Differentiate between AC motors and DC motors. [8]

OR

- Q6) a)** What are the factors to be considered while Selection and sizing of the motor? [8]
- b)** Explain different types of drives used in electric vehicle with neat sketches. [8]

- Q7) a)** Explain the significance of implementation of IOT in electric vehicle on basis of wireless sensor network with neat sketch wherever required. [10]
- b)** List out the customer related issues for use of electric vehicles? Also explain Need for IoT in EV? [8]

OR

- Q8) a)** Explain the Significance of IOT for modes of fast and efficient charging in electric vehicles. [8]
- b)** Write a short note on intelligent transport system with its advantages? [10]

